

Food Bank Client Proposal: Predicting Failed Food Businesses

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1. Entity

The Greater Vancouver Food Bank, or GVFB, is a real charity in Vancouver, BC. In 2025 it gave out around 9.4 million pounds of food across more than 440,000 visits. It also helps the community by supporting local food businesses like small restaurants and grocery stores. When a small business closes, the neighbourhood loses a cheap place to get food. That makes it harder for people in need to get free food from the food bank. GVFB wants to spot food businesses that look like they will fail, early, so it can help them before they close. Food businesses are partners that help GVFB do its work, so it is important for GVFB to look out for smaller businesses that help them too.

2. Problem Definition

We want a model that looks at a food business when it first gets its licence and predicts: will this business fail later. Failed means the licence gets cancelled or the business is marked gone out of business. The label is simple: failed = 1 if the status says 'Cancelled' or 'Gone Out of Business', and 0 if not. We can only use info we know when the licence is given, nothing after. The model gives a score from 0 to 1 per business so the food bank knows who to help first.

- Is there a pattern in the licence data that shows which food businesses will fail?
- Which early signs matter most, business type or staff count?
- Can we rank new businesses so the food bank helps the most at risk first?

This is a supervised prediction task because the answer is a clear yes or no. It is not a clustering task.

3. Stakeholder / Audience

The main person using this is the Community Food Access Coordinator at GVFB. They decide which new food businesses to support first. A ranked list, with the most at risk at the top, lets them spend limited help on the businesses most likely to fail.

- Coordinator: uses the risk score to pick which businesses to contact first.
- Planning lead: a second reader who wants to see if certain business types or neighbourhoods need special help.

The answer directly informs the decision of which newly licensed food businesses to onboard into support or surplus food partnerships each week.

4. Datasets

We use a real file from the City of Vancouver open data site, `vancouver_food_businesses_sample.csv`. It has 221 real food businesses and 12 columns. I downloaded the city licence data myself, kept only the food businesses, and built the failed label. You can pull the full source again from the city site anytime.

Food Business Licence Dimensions:

- licencenumber (ID): the licence identifier.

- businessname (Nominal): name of the business.
- businesstype (Nominal): what kind of food business, Restaurant, Food Wholesale, etc.
- localarea (Nominal): Vancouver neighbourhood, Downtown, Kitsilano, etc.
- status (Nominal): where the label comes from, Issued, Pending, Cancelled, Inactive, Gone Out of Business.
- failed (1/0): the label, 1 if Cancelled or Gone Out of Business, else 0.
- numberofemployees (Quantitative): staff reported at licence time, 1 to 4000.
- issueddate (Date): when the licence was given, about 88 percent filled.
- expireddate (Date): when the licence ends, about 88 percent filled.
- feepaid (Quantitative): licence fee paid in CAD, about 48 percent filled.
- folderyear (Ordinal): the year the application was filed.
- postalcode (Nominal): business postal code, almost all Vancouver, BC.

The failed label is already in the file. In this 221 business slice about 9 out of 100 are failed, so it is an uneven but workable yes or no task.

5. Success Criteria

A good answer gives the Coordinator a simple risk score for each food business, from 0 to 1. Good means three things:

- The model is better than a random guess at telling failed from not failed.
- It shows which early signs matter most, like business type or staff count, so the team knows what to act on.
- It gives a clear cutoff, like contact the 10 most at risk each week, so the prediction drives early help.

From the stakeholder view, success is a ranked list the Coordinator can act on the same week, not just a number in a report.

6. Constraints

Real data has gaps. Some columns are not filled in for every row, issueddate and expireddate about 88 percent, feepaid about 48 percent. Only use info we know when the licence is given, nothing after. This predicts risk, it does not prove why a business fails, and we cannot add outside data.

7. References

- City of Vancouver Open Data portal, Business Licences dataset (opendata.vancouver.ca).
- GVFB 2025 Impact Report (foodbank.bc.ca).
- Food Banks Canada 2025 HungerCount (foodbankscanada.ca).

Food business subset of 221 rows filtered and verified 2026-07-21. Vancouver open licence data is also used in IAT 461 Assignment 3, so it fits the course.